HARDWARE & MANUFACTURING SPECIFICATION · SKU PIA24028P

Octopus 16 Board Data, Starter-Board Ratings, Kit Test

Bare-board data for the PiEEG Octopus 16, the published ratings of the Elecrow starter board, and the test that checks a complete kit. Circuit diagrams, copper artwork and parts lists are not part of this document.



4-layer	Ø 26	328P	0.69 W	PP
Octopus 16 PCB	mm round board	ATmega starter MCU	Board, all modules on	Case material

01 OCTOPUS 16 — BARE-BOARD PARAMETERS

Parameter	Specification	Note
Base material	**FR-4**, εr 4.5	Standard Tg
Layers	**4**	1 oz copper on all layers
Dimensions	**Ø 26.0 mm**, 1.6 mm	Single round board
Solder mask / legend	**Black** / white	Both sides
Surface finish	**HASL** (production) · ENIG optional	ENIG gives flatter pogo-pin pads
Minimum track	0.15 mm (6 mil)	Vias Ø 0.30 mm drill / Ø 0.60 mm pad
Assembly	Double-sided SMT, no through-hole parts	18 pogo pins on the pin side

Full Octopus 16 data — stack-up, design rules, assembly — is in the Octopus 16 documentation set (OC16-PCB-003).

02 STARTER BOARD — PUBLISHED RATINGS (ELECROW SEA0004T)

CONTROLLER	
MCU	ATmega328P · 16 MHz crystal · 32 KB flash, 2 KB RAM, 1 KB EEPROM
USB	USB-C, CH340C USB-to-UART (RX/TX LEDs)
Power path	5 V DC-DC (VDD_5V_SYS) for 5 V modules; 3.3 V regulator for IMU, PIR and light sensor
ELECTRICAL	
Operating voltage	5 V
Input voltage	DC 5–20 V recommended (absolute 3.6–32 V)
Consumption	42 mA / 0.21 W minimum system · 138 mA / 0.69 W all functions on
LCD 1602	5 V · 1.7 mA logic, 24 mA backlight max · display area 64.5 × 14.0 mm
Crowtail ports	5 V, 20 mA per I/O pin
MECHANICAL & ENVIRONMENT	
Dimensions / weight	195 × 170 × 46 mm · 380 g
Case material	PP (polypropylene)
Storage	−40 to +85 °C (normal conditions)

Source: Elecrow "All-in-One Starter Kit for Arduino" datasheet V1.0 (2025-04-17), included in the OctoBio-kit repository. Board design and manufacture by Elecrow.

03 KIT QUALITY TEST

1 Octopus 16

Fit the XIAO (5V / GND / 3V3 aligned), flash Install Octopus-16 at octobridge.pieeg.com/flash; it must advertise as **Octopus-16-XXXX**, the LED must stay solid when connected, and all 16 channels must stream with a continuous counter.

2 Starter board

Flash **octoBio Kit** (about 20 s). The LCD shows "octoBio Kit Ready" and the board answers **INF protocol=1 ready** at 115200 baud.

3 Board page

With both connected: knob, light, distance and button change live; LED D10, buzzer, servo and LCD respond to the Board page controls.

4 Signal check

Octopus 16 on the head on battery power: "EEG ok" green (RMS 2–200 µV), alpha rises with the eyes closed, a blink raises RMS and gamma.

04 HANDLING AND COMPLIANCE

Topic	Requirement
ESD and storage	Handle both boards by the edges; keep the Octopus 16 pins protected from side loads; store the starter board in its case.
Pogo pins	Press straight down only. Clean with isopropyl alcohol > 90 %; never oil or grease.
Relay	Low-voltage loads only. Never connect it to mains voltage.
Classification	Learning and experimentation kit. Not a medical device; focus and relax values are uncalibrated.
Design files	Circuits, layouts and parts lists of both boards are not part of the user documentation.

05 RELATED DOCUMENTS

OB-DS-001 Product Datasheet	**OB-MD-002** Mechanical & Layout Drawing	**OB-PIN-004** Pinout & Connection Guide
OB-QSG-005 Quick Start & Safety Guide	**OB-SW-006** Software & Protocol Guide	